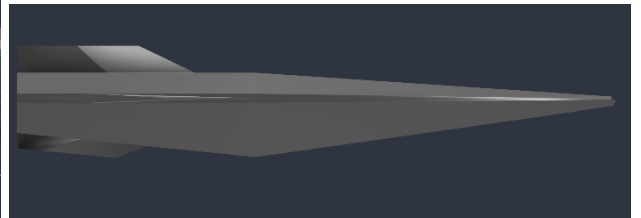
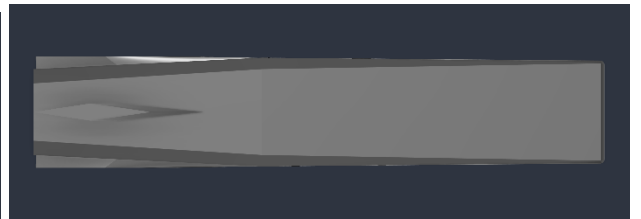
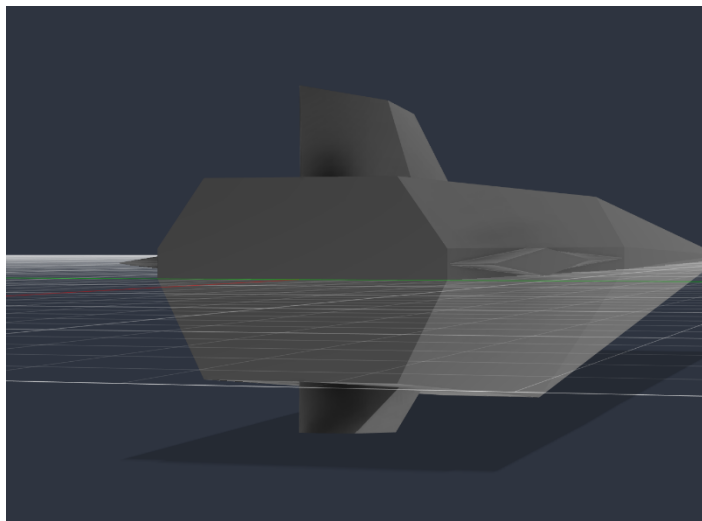
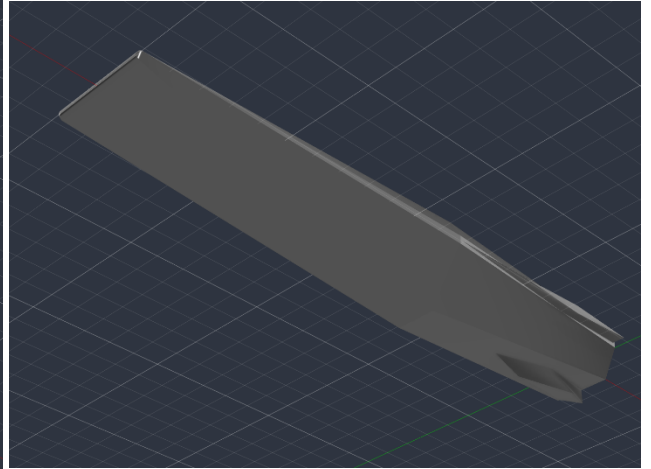
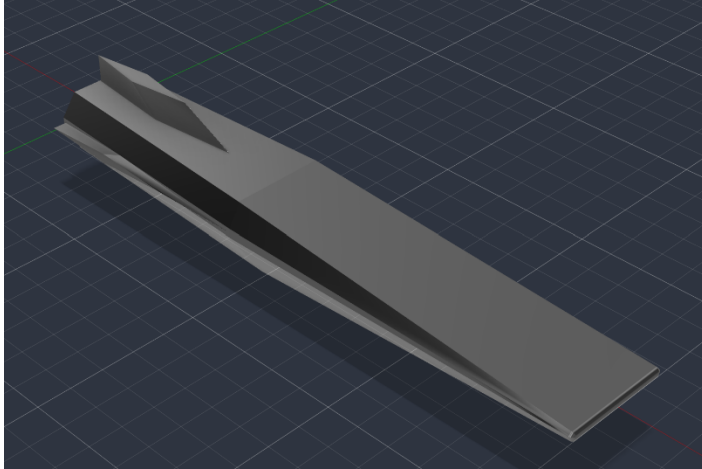


Mustang



**The design features a blunted nose for heating and various control surfaces.
There are few panels to maintain manufacturability.**

Inspirations

Based off the X-43 and the X-15.



X-43

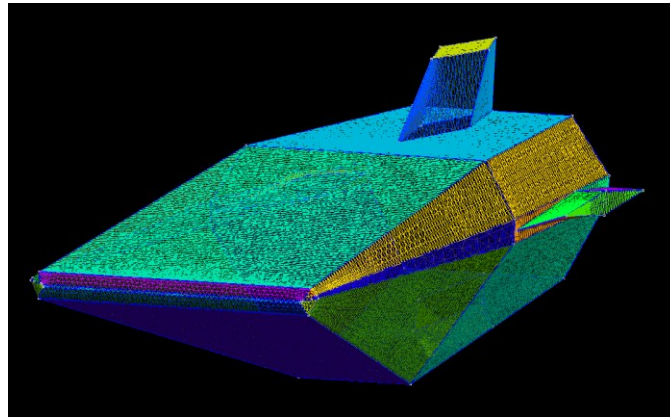


X-15

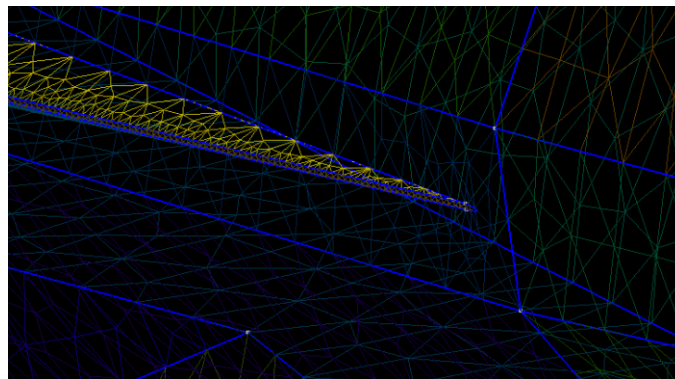
The aircraft uses elements from both. Namely, the wedge shape of the X-43 and the large control surfaces on the X-15.

Analysis

The following is the mesh used, proceeded by the CBAero results.



Front



Wing Tip

Parameters

Planform Area = 0.753 m^2

Wing Chord = 0.775 m

X = -0.8

Scale Factor = 0.001

Altitude = 25 km , with an air density of $0.395 \frac{\text{kg}}{\text{m}^3}$, and sea level.

Result Highlights

25 km

2.5773 is the highest LoD at Mach 4; 5 Alpha.

Sea Level

In the same conditions except at sea level the LoD was 2.677

Making the Reference Chord 1.775 instead of 0.775

Caused the LoD (sea level Mach 4; 5 Alpha) to change from 2.677 to 2.677

Design Changes

I will continue working on the design; I will make it more shallow and exaggerate the boattail and see what happens.